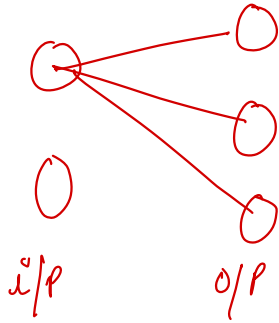


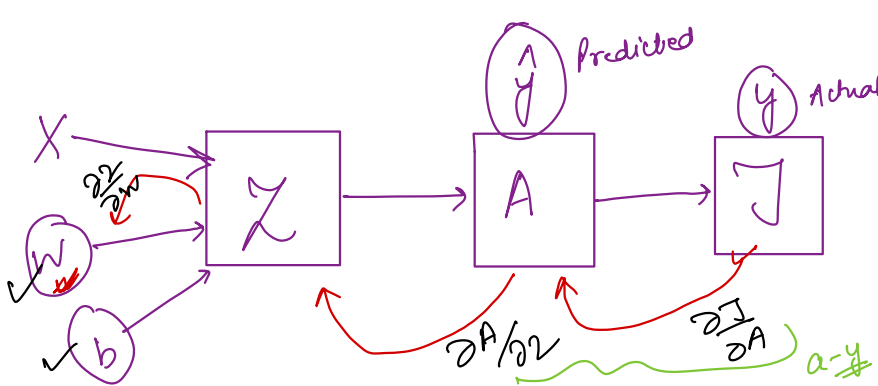
Agenda

→ Activation function

→ Equation → 1 hidden layer



What is hidden?
How do we write eq for
Backpropagation when
we have a hidden layer



$$A = \sigma(z)$$

$$\begin{matrix} 100 \downarrow \\ \frac{\partial J}{\partial w} \end{matrix}, \begin{matrix} \frac{\partial J}{\partial b} \end{matrix}$$

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial A} \times \frac{\partial A}{\partial z} \times \frac{\partial z}{\partial w}$$

Forward pass

$$z = Xw + b$$

$$A = \sigma(z)$$

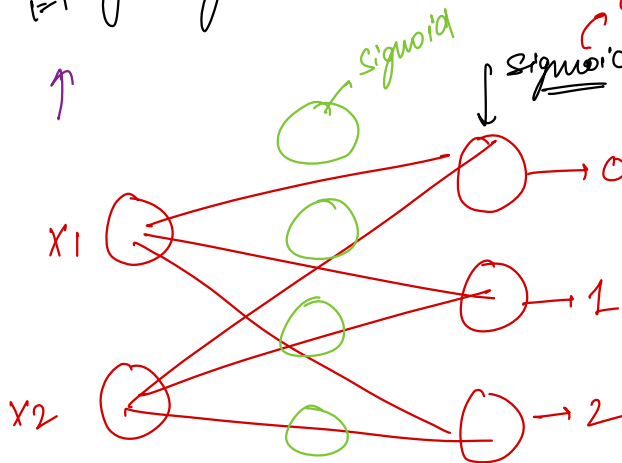
$$W^{\text{new}} = W^{\text{old}} - \eta \Delta W$$

learning rate.

$J \rightarrow$ cost function (multiclass)

$$= -\frac{1}{n} \sum_{i=1}^n y_i \log(A_i)$$

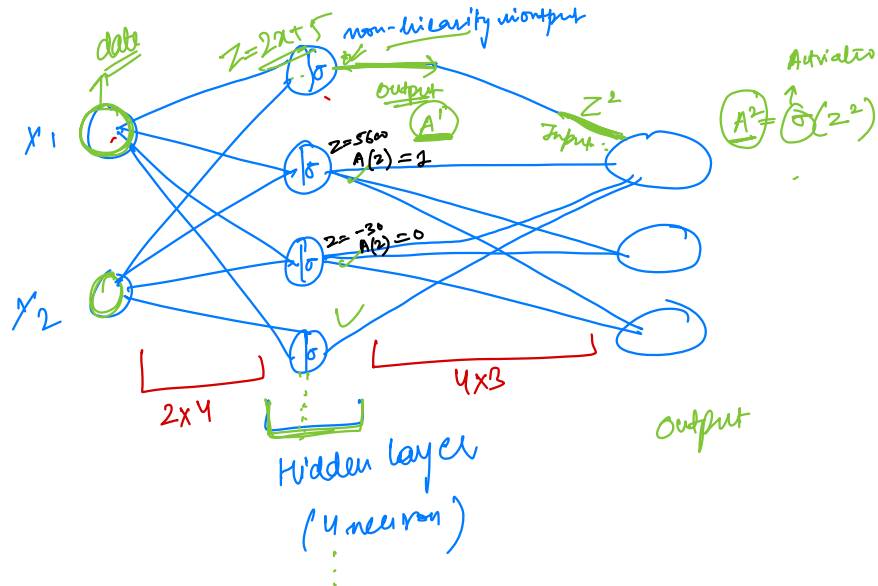
Coded this ↑

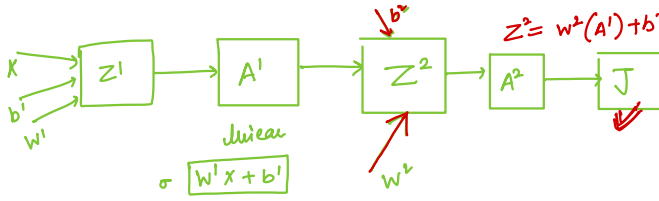


Currently, it is used

(non-linear function)

If we use sigmoid for feature engineer, non-linear feature.





$$Z^2 = W^2(A^1) + b^2$$

$$= W^2(\text{Act}(W^1x + b^1)) + b^2$$

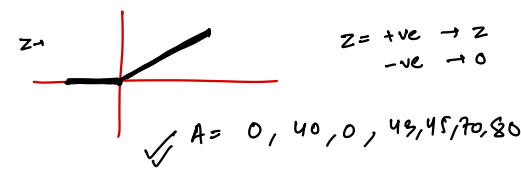
Not get a poly degree
 $\Rightarrow$ not get non-linearity
 $\Rightarrow$ use a non-linear function
 $\Rightarrow$ Sigmoid for non-linear output

Is Sigmoid the non-linear Act function

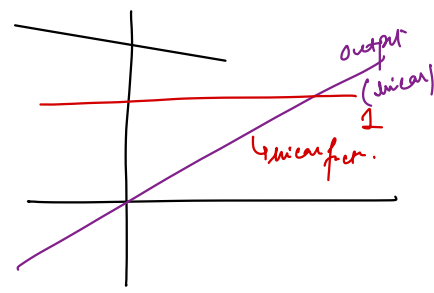
① Linear $A(z)$ = $\alpha z + \beta$
 = $[-1, 1]$

-ve pos $X \rightarrow$ non-linear activation

$Z = -36, 40, -20, 43, 45, 70, 80$



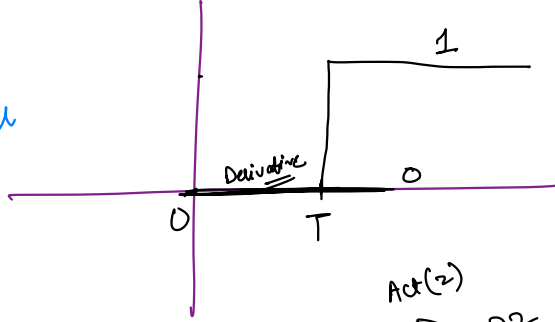
$\frac{df(x)}{dx} = 1$ $f(x) = x$



② step function

(HW)

→ Output layer
→ basic ~~layer~~



$$A(z) = \begin{cases} 1 & z > T \\ 0 & z < T \end{cases}$$

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial A} \times \frac{\partial A}{\partial z} \times \frac{\partial z}{\partial w}$$

Below the equation, a circle contains $\frac{\partial A}{\partial z}$ with an arrow pointing down to 0. A pink dotted line extends from the right side of the equation.

$$w_{new} = w_{old} - \alpha \frac{\partial J}{\partial w} \rightarrow 0$$

$$\frac{\partial J}{\partial w} = 0$$

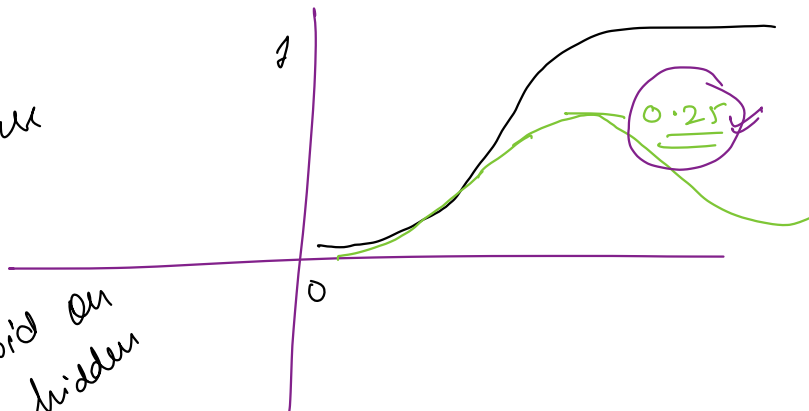
Vanishing gradient problem

② Sigmoid

What if there are

50 hidden
layer.

Vse Sigmoid on
all the hidden
layer.



$$Z = -\infty, \quad \boxed{[0, 1]}$$

$$= +\infty$$

Continuous

Reprocessing $\therefore \rightarrow$ 50 layer



$$0.25 \times 0.25 \times 0.25 \dots \rightarrow \boxed{\approx 0}$$

Variety
grad
prob

$[0, 1]$

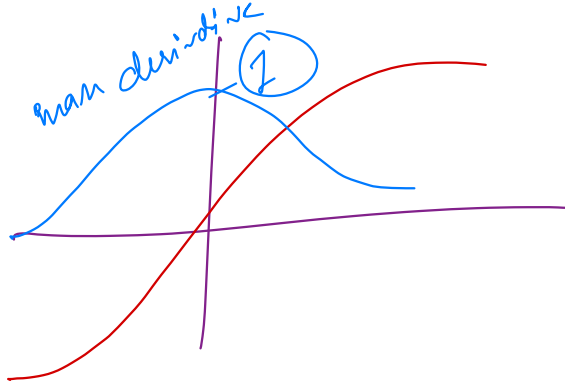
Compute expensive

④ softmax (hidden) $\rightarrow$ loss of info propagation

$$\begin{bmatrix} 0.9998 \\ 0.002 \end{bmatrix} \rightarrow \text{no}$$

⑤ tanh $\rightarrow$ like Sigmoid, it includes -ve values as well

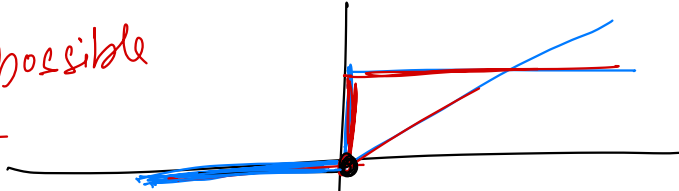
$$[-1, 1]$$



$$A(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

⑥ ReLU $\rightarrow$ Rectified Linear Unit.

$\rightarrow$ easiest possible derivative



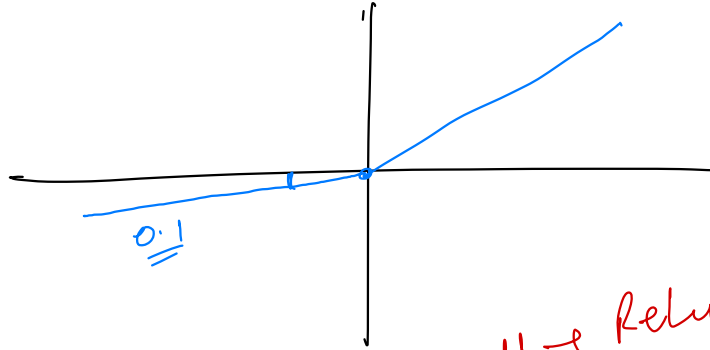
$\rightarrow$ only 1 bend
 $\rightarrow$ faster to compute
 $\rightarrow$ min possible non-linearity

$$A(z) = \begin{cases} z & ; z > 0 \\ 0 & ; z \leq 0 \end{cases}$$

$$\max(0, z)$$

$\rightarrow$ could be some problem of vanishing gradients.

7) Leaky Relu

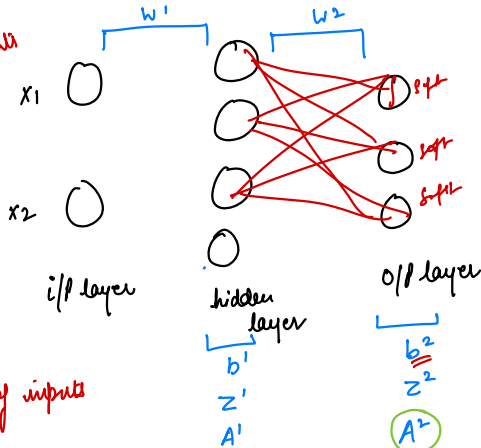


fast learning | network is small $\rightarrow$ Relu $\rightarrow$
otherwise $\rightarrow$ leaky

$$\begin{aligned} z > T &\rightarrow \overset{\text{gen}}{\text{0}} \\ z < T &\rightarrow \text{output } \textcircled{z} \\ &\quad 0.1z \end{aligned}$$

learning will be
slower than
relu

300 data points



d → no. of inputs

Shape

$$X = 300 \times 2$$

$$m \times d$$

$$w^1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \times 4$$

$$b^1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \times 4$$

$$z^1 = w^1 \cdot X + b^1$$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 300 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$z^1 \Rightarrow \begin{pmatrix} 300 \\ 4 \end{pmatrix}$$

$$A^1 = \begin{pmatrix} 300 \\ 4 \end{pmatrix}$$

$$w^2 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$z^2 = A^1 \cdot w^2 + b^2$$

$$\Rightarrow \begin{pmatrix} 300 \\ 4 \end{pmatrix} \times \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 300 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 300 \\ 3 \end{pmatrix}$$

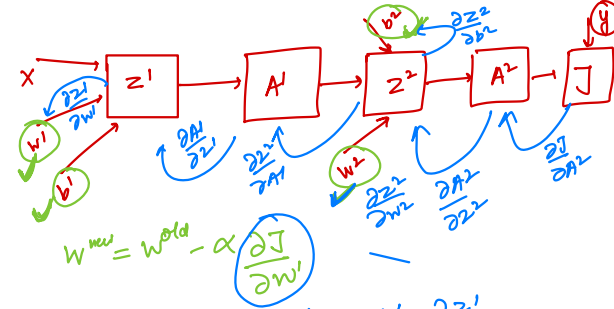
$$A^2 = \begin{pmatrix} 300 \\ 3 \end{pmatrix}$$

x_1	x_2
2	3

Class 1

1	1	1
0	1	2
0	1	0

$$A^2 = \begin{pmatrix} 300 \times 3 \end{pmatrix}$$



$$w_{new} = w_{old} - \alpha \frac{\partial J}{\partial w_1}$$

$$\frac{\partial J}{\partial w_1} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial z^2} \times \frac{\partial z^2}{\partial A^1} \times \frac{\partial A^1}{\partial z^1} \times \frac{\partial z^1}{\partial w_1}$$

$$\frac{\partial J}{\partial w_2} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial z^2} \times \frac{\partial z^2}{\partial w_2}$$

$$z^2 = w^2 A^1 + b^2$$

$$\frac{\partial z^2}{\partial w^2} = A^1$$

$$\Delta w^2 = \text{pred} - \text{Actuals}$$

$$\frac{\partial J}{\partial w_2} = (A^2 - y) \cdot A^1$$

$$\frac{\partial J}{\partial b^2} = A^2 - y$$

$$\Delta w^2_{\text{slope}} = (A^2 - y) \cdot (A^1)$$

$$= A^1 \cdot T(A^2 - y)$$

$$= (3w, 4) \cdot (3w, 3)$$

$$\rightarrow (4, 3w) \cdot (3w, 3)$$

$$\rightarrow (4, 3) \quad \cancel{\rightarrow (3, 4)}$$